Positive-measure avoidance with growing logarithmic gaps

Yongxian Zhang*

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Abstract

We prove positive-measure affine non-universality for decreasing null sequences whose consecutive logarithmic gaps are $o(\log\log z)$, where $z = -\log_2 a$. This includes sequences whose adjacent ratios tend to zero and whose occupied logarithmic bins have upper Banach density zero. A more general theorem needs this filling condition only on arbitrarily late multiplicative logarithmic windows. For each prescribed countable family we construct a closed periodic set of measure arbitrarily close to one that misses infinitely many distinct outputs in every tail of every nonzero leading-power map with a power-controlled remainder. The proof allows the branching, depth and separation gap of a finite routing tree to grow with its position. Its new schedule has total span $U^{o(1)}$, which leaves sufficient room for the error buffer.

The finite routing construction is inherited and explicitly reproved below. The contributions here are the variable-parameter schedule, local-window sampling, the resulting zero-density sequence classes and the intermittent window extension. No complete solution of the Erdős similarity conjecture, world-first claim or external human peer review is asserted.

1. Problem, notation and statements

All unmarked logarithms are natural. Lebesgue measure is denoted by m. For a measurable one-periodic set S, put $\operatorname{dens}(S) = m(S \cap [0,1))$. A positive null sequence is strictly decreasing, $a_n > 0$, $a_n \to 0$. Discarding finitely many initial terms will always be permitted when selecting test inputs, and never weakens the eventual avoidance conclusion. Put $z_n = -\log_2 a_n$; eventually $z_n > e$, and z_n increases to infinity.

An affine copy of $A = \{a_n\}$ is $y + cA$ with $c \neq 0$. The Erdős similarity conjecture asks whether every infinite subset of the line is absent in this sense from some positive-measure measurable set. We treat a specified class of null sequences, not all infinite sets.

Definition 1 (filling on late annuli). A sequence (z_n) has property W if, for every integer $R \geq 2$, there are integers $U_j \to \infty$ and real numbers $D_j \geq 1$ with

$$\frac{D_j}{\log\log U_j} \longrightarrow 0, \tag{1}$$

*School of Computer Science and Engineering, South China University of Technology, Guangzhou, China. Email: mxymmxym1@gmail.com. ORCID: 0009-0000-3864-3536.

such that every closed interval $[v, v + D_j]$ contained in $[U_j/R, RU_j]$ contains a member of (z_n). Only sufficiently large j are relevant, so all logarithms in (1) have positive arguments.

Theorem 2 (local-window avoidance). Let (A_l) be a prescribed nonempty countable family of positive null sequences whose logarithms have property W. For every $0 < \varepsilon < 1$ there is a closed nowhere dense one-periodic set $E \subset \mathbb{R}$ with $m(E \cap [t, t + 1]) > 1 - \varepsilon$ for every real t, with the following property. After E has been chosen, fix any family member and any $s, \alpha > 0$, $c \neq 0$, $y \in \mathbb{R}$, and $M \geq 0$. If f is defined on a sufficiently small tail of that member and

$$|f(a) - y - ca^s| \leq Ma^{s+\alpha} \tag{2}$$

there, then in every positive input tail there are infinitely many distinct values of $f(a)$ outside E. In particular $F = E \cap [0, 1]$ is compact, $m(F) > 1 - \varepsilon$, and contains no nontrivial affine copy of any family member.

The family is fixed before E; the theorem does not construct a single set for all null sequences. There are no regularity or injectivity assumptions on f beyond (2). Its domain need only be the stated tail.

Corollary 3 (growing consecutive gaps). Theorem 2 applies whenever

$$z_{n+1} - z_n = o(\log \log z_n). \tag{3}$$

Corollary 4 (ratios tending to zero). For every fixed $0 < \beta < 1$ it applies to

$$a_n = 2^{-n(\log \log(n+20))^\beta}, \qquad n \geq 1. \tag{4}$$

For this sequence $a_{n+1}/a_n \to 0$, and its occupied dyadic logarithmic bins have upper Banach density zero. One may also prescribe countably many such sequences at once. The assertion is not simultaneous over an unspecified uncountable family of β.

2. Sampling from a finite logarithmic annulus

Fix one sequence, a requested input tail $a < 2^{-h}$, and a compact exponent rectangle $0 < s_0 \leq s_1$. Fix an integer coefficient scale k. Write

$$\delta = 4/s_0, \qquad R \geq \max(2, 4s_1, 8/s_0), \tag{5}$$

with integer R. Choose the annuli from Definition 1 for this R. All subsequent choices are made separately on the jth annulus.

Start with a logarithm in $[U_j/R, U_j/R + D_j]$, which exists for large j. Thereafter select greedily the first original logarithm strictly larger than the previous selected one plus δ. Continue indefinitely, not just inside the annulus. Denote the selected logarithms by ζ_i and the inputs by $b_i = 2^{-\zeta_i}$. For large j every selected input is in the requested tail. Consecutive selected logarithms have gaps greater than δ.

Whenever $\zeta_i + \delta + 1/2 + D_j \leq RU_j$, the interval $[\zeta_i + \delta + 1/2, \zeta_i + \delta + 1/2 + D_j]$ contains an original logarithm. The first eligible logarithm therefore satisfies

$$\delta < \zeta_{i+1} - \zeta_i < \delta + D_j + 1. \tag{6}$$

Put

$$D = s_1(\delta + D_j + 2), \quad \eta = 1/(2D). \tag{7}$$

In particular $D \geq 4$, and $D = o(\log\log U_j)$ along these annuli. Suppose a routing template lies in $[U, U+T]$, with $U = U_j$ and $T \leq U$. For all sufficiently large j, its entire relevant logarithm range, including the extra $\delta + D_j + 1$, lies strictly below RU. The first selected logarithm lies below $(U+k)/s_1$; these assertions follow from (5), $D_j = o(U)$, and the fact that k, δ are fixed.

For each $s \in [s_0, s_1]$, set $w_i = s\zeta_i - k$. The sequence starts below U, is separated by more than three, and every gap needed to cross a window inside $[U, U+T]$ is less than D, by (6). Thus for every open window $(u, u+\ell)$ in that template, if $\ell \geq 2D$,

$$\eta\ell \leq \#\{i : u < w_i < u + \ell\} \leq 1 + \ell/3. \tag{8}$$

For the lower bound, the last member at or below u and the first member at or above $u+\ell$ enclose r strictly interior members. The length is at most $(r+1)D$, so $r \geq \ell/D - 1 \geq \ell/(2D)$. This argument includes equality at either endpoint. The upper bound uses separation. None of (6)–(8) requires a bound on the logarithmic gaps after leaving the annulus.

3. A uniform finite routing lemma

Here we give the finite construction with all its numerical guards. They are valid for each parameter choice separately; their proof never requires the tree parameters to be fixed while U tends to infinity.

Fix $0 < p < 1/12$. Choose integers $b \geq 2$, $d \geq 1$, $g \geq 1$, and $L \geq 2$. Use a complete ordered b-ary tree of depth d, with $K = \sum_{r=1}^{d} b^r$ edges. Recursively define

$$T_1 = bL + (b-1)g, \qquad L_r = g + T_{r-1}, \quad T_r = 2bT_{r-1} + (3b-1)g \quad (r \geq 2). \tag{9}$$

Place an integer window of length L_r on each edge from a height-r vertex. The edge is followed by its whole child subtree, and consecutive pieces are separated by g unused integers. The template starts at U. Its total span is $T = T_d$. An edge window of length $\ell = L_r$, together with its subtree, has span at most 2ℓ. All lengths are at least L. These facts follow directly from (9), including the height-one case. More explicitly, the preorder traversal places g unused indices between every consecutive pair of edge windows. A window starting at u ends at $v = u + \ell - 1$; the last endpoint of the whole template is $U + T - 1$.

Assume the local samples obey (8), and the following guards hold:

$$\begin{gathered} U \geq 4, \quad U > |k|, \quad T \leq U, \quad L \geq 2D, \quad L \geq g, \qquad \lambda = p\eta(b-1)/2, \quad \kappa = \lambda - 4\log 2 > 0, \\ (1 - 2^{1-b})^d < p, \qquad K2^{3-g} < p, \\ 46080b^2(U+1)^2 e^{-\kappa L} < p. \end{gathered} \tag{10}$$

The initially selected sample must precede U in every output exponent, as proved in Section 2. Finally fix an error exponent $\alpha_0 > 0$ and an integer error bound $q \geq 1$. Let

$$N = 2^{U+T+2}, \quad r_U = (q2^{-k} + 1)2^{-U}2^{-\alpha_0(U+k)/s_1}, \quad 4Nr_U < p. \tag{11}$$

Lemma 5 (finite blocker). Under these assumptions there is an open one-periodic set H of density less than $6p$ such that, for every $x \in \mathbb{R}$, $s \in [s_0, s_1]$, $t \in [1, 2]$ and arbitrary errors $|e(b_i)| \leq qb_i^{s+\alpha_0}$, at least one selected input satisfies

$$x + t2^k b_i^s + e(b_i) \in H. \tag{12}$$

The index in (12) may lie beyond the finite template. This is essential for repairing exceptional centers.

3.1 Tables, stable centers and actual addresses

At integer endpoint v use the periodic grid of 2^{v+3} half-open cells. The address of a real point is $\lfloor 2^{v+3}x \rfloor \bmod 2^{v+3}$, and cells are left closed and right open. At each nonlast outgoing edge put independent fair bits indexed by its endpoint-grid cells. At each leaf put independent Bernoulli-p bits indexed by its incoming endpoint grid. Entries of different tables are independent. Route a point to the first nonlast child whose bit is one, or to the last child when all those bits are zero. Include the point in B exactly when its reached terminal bit is one. The set B is a union of cells on the finest N-grid. For a fixed point, conditioning on the selector tables leaves its terminal entry Bernoulli-p; hence $\mathbb{E}\ \mathrm{dens}(B) = p$.

If a window begins at $u = v + g + 1$ after a preceding endpoint v, call a center unstable when a boundary of the 2^{v+3} grid lies in $(x, x + 2^{-v-g}]$. Over one period its measure is at most 2^{3-g}. The unstable union has measure at most $K2^{3-g} < p$.

An active offset on a length-ℓ window obeys

$$2^{-u-\ell} < t2^{k-s\zeta_i} < 2^{1-u}. \tag{13}$$

At a stable center it shares all earlier grid addresses with the center, since (13) is smaller than the stability allowance. At its own grid its address differs from the center's; the grid width is one quarter of the lower bound in (13). Different active inputs have offsets in a ratio greater than eight, so their addresses differ too. Offsets lie in an interval of length less than $1/8$; periodic wrapping cannot defeat these separations. The distinctions persist on every finer grid. Exact grid boundaries use the stated half-open convention.

3.2 Conditional probability at a first default

At a stable center expose its addressed entry in every selector table. This determines its entire route. The probability that its route has no default child is exactly $(1 - 2^{1-b})^d < p$. Otherwise condition on the complete exposure atom and its first default, at height r.

For each of the $b - 1$ nondefault edges from that vertex and each active input, test its own selector bit and then the terminal bit found by routing it in that child subtree. Earlier addresses agree with

the center; the required preceding nondefault bits are zero. A successful test is therefore a point of the actual global set B, not of a hypothetical local set.

There are at least $m_* \geq \eta(b-1)L_r$ tests. Their own selector entries are distinct and were not exposed. Condition next on all selectors. Points that reach different leaves use different terminal tables. Points that reach the same leaf have distinct addresses already on their common own edge, and hence distinct finer terminal addresses. Consequently the conditional probability of failure of all tests is $(1-p)^{\sum A_i}$, where A_i are the tested own selector bits. Averaging the independent fair A_i gives

$$(1-p/2)^{m_*} \leq e^{-\lambda L_r}. \tag{14}$$

Shared auxiliary selector reads do not affect this calculation. Entire routing paths are not being assumed independent. Multiplying (14) by the exposure-atom mass yields a joint estimate, also valid for zero-mass atoms.

3.3 The continuum of parameters and all boundary strata

Fix the center and the first-default vertex, and put $\ell = L_r$. There are at most

$$P = (b-1)\left(\lceil (U+T+|k|)/3 \rceil + 1\right) \tag{15}$$

candidate edge–input pairs: any candidate logarithm is positive, is below $(U+T+|k|)/s_0$, and is separated from the others by more than $3/s_0$. For each pair retain the two activation equations. Retain also all boundaries of its local finest grid whose positive offsets from the center are less than 2^{1-u}. Its local span is at most 2ℓ, so there are at most $1+2^{2\ell+3}$ such boundaries.

Writing $v = \log t$, a boundary of positive offset β is crossed at

$$v - s\zeta_i \log 2 + k \log 2 - \log \beta = 0. \tag{16}$$

All retained equations are affine lines in (s, v). Their full sign vectors determine activation and each actual local finest-grid address. Inactive values at activation equality and right-cell floor values at grid equality are encoded by zero signs. At most $m = P(3+2^{2\ell+3})$ lines occur.

For m affine lines in the plane the total number of distinct sign vectors is at most $2m^2+1$: after merging coincident lines, there are at most $m(m-1)/2$ vertices, m^2 open line edges, and $1+m(m+1)/2$ faces. Each stratum has a constant full sign vector. This count also handles parallel lines, repeated intersections and constant equations. Intersect the strata with the closed parameter rectangle and select one point from each realized sign vector. No table entries are used in that choice.

The resulting representative count is bounded by $20(m+5)^2$. Union bounding (14) over these representatives gives a continuum miss probability on the exposure atom at most its mass times

$$5120b^2(U+T+|k|+2)^2 e^{-\kappa\ell} \leq 46080 b^2 (U+1)^2 e^{-\kappa L} < p. \tag{17}$$

Indeed $P \leq b(U+T+|k|+2)$ and, for $P \geq 1, \ell \geq 2$, $P(3+2^{2\ell+3})+5 \leq 16P2^{2\ell}$. Squaring accounts for $4\log 2$ in κ; $T \leq U$ and $|k| < U$ account for the factor nine in (17). The representatives work

for every assignment of unexposed bits. They need not be selected measurably as functions of the center; only a pointwise bound is used before a finite sum and integration.

3.4 Closed exceptional centers and open repair

Let $B_1 = B + (-r_U, r_U)$ and $B_2 = B + (-2r_U, 2r_U)$. Enlarging a union of finest-grid cells by radius $2r_U$ increases its periodic density by at most $4Nr_U$. Therefore $\mathbb{E}\,\mathrm{dens}(B_2) < 2p$. There are finitely many potentially active inputs: their logarithms are bounded above by $(U + T + |k|)/s_0$. Let I contain them, and let S_i be the relatively open set of exponents for which $s\zeta_i - k$ belongs to the union of all the open template windows. Define Q as the set of centers for which some parameter pair has, for every $i \in I$, either $s \notin S_i$ or its unperturbed output outside B_1.

Each inactivity condition is closed in the closed parameter rectangle, because activation windows are open. Each outside-B_1 condition is closed by continuity. Their finite intersection is closed, and projection along the compact rectangle is closed. Thus Q is closed and periodic. At stable centers, membership forces a continuum miss at the first default or a route with no default. Equations (10) and (17) bound its probability by $2p$. Integrating and allowing the unstable union gives $\mathbb{E}\,\mathrm{dens}(Q) \le 3p$. This uses only finite sums and measurable indicators.

Choose an outcome with $\mathrm{dens}(B_2) + \mathrm{dens}(Q) \le 5p$. Thicken the closed periodic Q to an open periodic V with $\mathrm{dens}(V) < \mathrm{dens}(Q) + p$; continuity from above on a finite period justifies this, including the empty-set case. Put $H = B_2 \cup V$. Its density is less than $6p$.

If $x \notin Q$, some active input has unperturbed output in B_1. Activation gives $s\zeta_i - k > U$, so its allowed error satisfies

$$qb_i^{s+\alpha_0} \le q2^{-k}2^{-U}2^{-\alpha_0(U+k)/s_1} < r_U. \tag{18}$$

That perturbed output belongs to B_2. If $x \in Q \subset V$, use the full selected tail: the perturbed offsets converge to zero uniformly under the fixed bounds $2^{k+1}b_i^{s_0} + qb_i^{s_0+\alpha_0}$. An output is eventually in the open neighborhood V of x. This input need not be active in the finite template. These two cases prove Lemma 5.

4. Growing trees fit on the same annulus

We now prove the main new scheduling step. In this section all tree parameters are allowed to depend on j, equivalently on $U = U_j$. They are chosen before any table outcome or center.

Take D from (7), and set

$$\begin{aligned} b &= 2 + \lceil 24D/p \rceil, \qquad d = \left\lceil 2^{b-1}\log(2/p) \right\rceil, \\ K &= \sum_{r=1}^{d} b^r, \qquad g = 1 + \lceil \log_2(8K/p) \rceil, \\ \lambda &= p(b-1)/(4D), \qquad \kappa = \lambda - 4\log 2, \\ L &= \max\left(g, \lceil 2D \rceil, 2, \left\lceil \frac{4\log(U+2) + \log(46080b^2/p) + 1}{\kappa} \right\rceil \right). \end{aligned} \tag{19}$$

These are finite integers, however enormous. Since $\lambda \ge 6$ and $\log 2 < 3/4$, $\kappa > 3$. Also

$$(1-2^{1-b})^d \le e^{-2^{1-b}d} \le p/2 < p, \qquad K2^{3-g} < p. \tag{20}$$

The definition of L gives the last entropy guard directly:

$$46080b^2(U+1)^2e^{-\kappa L} \le pe^{-1}(U+1)^2/(U+2)^4 < p. \tag{21}$$

Lemma 6 (uniform tree-span bound). For every real $b \ge 2, g, L \ge 0$ and integer $r \ge 1$, the recurrence (9) satisfies

$$T_r + 2g \le b(2b)^{r-1}(L+2g), \qquad T_r \le (2b)^r(L+2g). \tag{22}$$

Proof. At $r=1$, $T_1 + 2g = bL + (b+1)g \le b(L+2g)$. For $r \ge 2$, $T_r + 2g = 2bT_{r-1} + (3b+1)g \le 2b(T_{r-1}+2g)$. Induction proves the first inequality. The second follows from $g \ge 0$, $L + 2g \ge 0$, and $b \le 2b$. This proof is uniform in depth and branching. □

Lemma 7 (sublinear span). With (19) and $D = o(\log\log U)$,

$$L = O_p(\log U), \qquad T = T_d = U^{o(1)}, \qquad T/U \to 0. \tag{23}$$

Proof. Put $t = \log U$, so $D = o(\log t)$ and $b = o(\log t)$. For every fixed $\theta > 0$,

$$d\log(2b) = t^{o(1)} = o(t). \tag{24}$$

Here is a direct justification, also covering bounded b. We have $d \le 1 + 2^{b-1}\log(2/p)$. The logarithm of the right-hand side times $\log(2b)$ is at most $b\log 2 + O_p(1) + \log\log(2b) = o(\log t)$. Thus the expression is eventually at most t^θ; take $\theta < 1$. The geometric sum gives $K \le 2b^d$, and hence $g = O_p(1 + d\log(2b)) = t^{o(1)} = o(t)$. The lower-bound terms $g, 2D, 2$ in (19) are $o(t)$ or bounded. The numerator of its final term is $4t + o(t) + O_p(1)$ and $\kappa > 3$, so $L = O_p(t)$. By (22),

$$\log T \le d\log(2b) + \log(L+2g) = o(t).$$

Consequently for every $\theta > 0$, $T \le U^\theta$ eventually. In particular $T/U \to 0$. Unlike a fixed-tree asymptotic argument, this calculation retains the changing depth and all changing coefficients. □

For large j we have $T \le U$, all the annulus margins in Section 2, and every standing integer guard. These statements follow along the very same chosen annuli; no independent choice of a later U with an older gap bound is made. Finally (11) obeys

$$4Nr_U = 2^{T+4}(q2^{-k}+1)2^{-\alpha_0(U+k)/s_1} \to 0, \tag{25}$$

because its logarithm is $-(\alpha_0\log 2/s_1)U + o(U) + O(1)$. Thus one sufficiently late annulus supplies all guards of Lemma 5. This proves a blocker of density less than $6p$ for every prescribed tail, bounded exponent rectangle, coefficient scale and error bound.

5. Completion of the global theorem

Enumerate the countable tuples (l, N, j, k, q, h), where $N \geq 2$, $j, q, h \geq 1$ are integers and $k \in \mathbb{Z}$. Assign strictly positive budgets $p_\tau < 1/12$ with $12 \sum_\tau p_\tau < \varepsilon$. Apply the blocker just proved with $[s_0, s_1] = [1/N, N]$, $\alpha_0 = 1/j$, the requested sequence and input tail, and coefficient scale k. Let its open periodic set be H_τ. Define

$$\mathcal{U} = \bigcup_\tau (H_\tau \cup (-H_\tau)), \qquad E = \mathbb{R} \setminus \mathcal{U}. \tag{26}$$

Then E is closed and periodic and its measure in a period is greater than $1 - \varepsilon$. For a measurable one-periodic set, splitting any unit interval at integers and translating its two pieces shows the same measure for every unit interval. Endpoints have measure zero.

Fix the data in (2). Choose N with $s \in [1/N, N]$, j with $1/j \leq \alpha$, and integer $q \geq \max(1, M)$. For $a < 1$ its remainder is at most $qa^{s+1/j}$. When $c > 0$, write $c = 2^k t$, $1 \leq t < 2$. For every sufficiently large h, the selected inputs are inside the actual domain of f and the eventual remainder bound. Lemma 5 at the tuple with that h produces a value in $H_\tau \subset \mathcal{U}$. When $c < 0$, apply it to $-f$ and then reflect; the value belongs to $-H_\tau$.

We therefore obtain missed values at inputs tending to zero. Eventually $(1/2)|c|a^s \leq |f(a) - y| \leq (3/2)|c|a^s$, so these values are different from y and tend to it. A finite set avoiding y cannot have that property. They give infinitely many distinct missed values inside every input tail. No injectivity was used. If E had an open interval, take an interior center and the map $f(a) = y + a$ on a small tail of any one family member; all outputs would be in that interval, a contradiction. Nonemptiness of the family is used here. Thus E is nowhere dense. Intersecting with $[0, 1]$ proves the compact assertion. This finishes Theorem 2. □

6. Consecutive gaps, explicit zero-density examples, and intermittent windows

Proof of Corollary 3. Discard a finite prefix so the logarithms are positive. Define $\Omega(Z) = \max\{z_{n+1} - z_n : z_n \leq Z\}$, using zero for an empty maximum. There are only finitely many entries under any finite Z. Hypothesis (3) implies $\Omega(Z) = o(\log\log Z)$: for any positive η, all sufficiently late entries are at most $\eta \log\log z_n \leq \eta \log\log Z$; the finitely many early entries contribute a fixed bound, whose ratio tends to zero. For a fixed integer R, set $U_j = j$ and $D_j = 1 + \Omega(Rj)$ once j is large. In any closed interval $[v, v + D_j] \subset [j/R, Rj]$, the last logarithm at or below v is followed by one at distance at most $\Omega(Rj) < D_j$ from it. Unless the left endpoint already is a logarithm, this successor is strictly inside the interval. Thus Definition 1 holds. Since $\log\log(Rj)/\log\log j \to 1$, its ratio condition holds too. □

Proof of Corollary 4. Put $\ell(x) = \log\log(x + 20)$ and $z(x) = x\ell(x)^\beta$. Then

$$z'(x) = \ell(x)^\beta + \frac{\beta x \ell(x)^{\beta-1}}{(x + 20)\log(x + 20)} \sim (\log\log x)^\beta. \tag{27}$$

The mean-value theorem gives $z_{n+1} - z_n \sim (\log\log n)^\beta$, while $\log\log z_n \sim \log\log n$. Their ratio tends to zero because $\beta < 1$; the gap itself tends to infinity because $\beta > 0$. Thus adjacent ratios tend to zero. The integer parts of the z_n are eventually distinct and have gaps tending to infinity. For each integer M, discard finitely many bins until these gaps are at least M. Every interval of L consecutive bins then contains at most $C_M + 1 + L/M$ occupied bins. Taking its supremum, then lim sup divided

by L, and then $M \to \infty$, proves upper Banach density zero. The existing positive-density criterion does not apply. $\square$

Property W is strictly broader than (3). Here is an explicit example. For integers $j \geq 2$, let $U_j = 2^{2^{j^2}}$ and take all logarithms

$$Z_j = \{\lceil U_j/j \rceil + mj : m \geq 0,\ \lceil U_j/j \rceil + mj \leq jU_j\}. \tag{28}$$

List the disjoint blocks Z_j in increasing order and set $a_n = 2^{-z_n}$. For each fixed R, all sufficiently large $j > 2R$ have $[U_j/R, RU_j]$ strictly inside the span of Z_j, with margins greater than $2j$. Every length-$2j$ subinterval there contains a lattice point. Also $2j/\log\log U_j \to 0$, since $\log\log U_j = j^2 \log 2 + \log\log 2$. Thus property W holds. The gap between successive blocks is far larger than $\log\log z_n$, so (3) fails. Within a late block gaps equal j, and between blocks they are larger; the same bin-counting argument proves zero upper Banach density. Theorem 2 therefore also treats sequences with arbitrarily large gaps outside their useful annuli.

7. Prior work, proof dependence and limitations

The finite center-exposure routing, parameter sign representatives and open repair used in Section 3 were developed in the geometric/continuum power-avoidance work publicly recorded in `mxym/math`, itself using the geometric-case method of OpenAI Mathematics [3]. The earlier theorem [4] assumed globally bounded occupied logarithmic gaps. Its complete release was replayed from all 65 owned source modules, with 568 public theorems and 35,620 safe-closure declarations checked in empty Lean kernel environments. The relevant published files and hashes are listed in `SOURCES.json`. We do not substitute that formalized bounded-gap endpoint for the new claim: Sections 2 and 4 are the added arguments and have complete written proofs. The full stated endpoint is also replayed in Lean: the variable-parameter late-window bridge and the countable-family Theorem 2 entry point are included in the accompanying formalization and checked from an empty kernel.

Cruz–Lai–Pramanik, *Large Sets Avoiding Affine Copies of Infinite Sequences*, Real Analysis Exchange 48(2) (2023), 251–270, arXiv:2204.12720 [1], construct a dimension-one Borel avoiding set under a slow-growth condition on $\log\log(a_n/a_{n+1})$. Its positive measure is not established; Section 4.4 explicitly discusses this missing point. Their dimension-one subset O has measure zero; this does not mean that their whole avoiding set E_K is proved to have measure zero. Our specified examples have positive-measure avoiding sets, indeed measure arbitrarily close to one. This is a distinction in the mathematical conclusions, not a certified priority assertion based on an exhaustive literature survey.

Feng–Lai–Xiong [2], prove bi-Lipschitz universality for suitable lacunary sequences, including (4). That uses transformations more general than affine maps and does not contradict this result. The earlier geometric case and the repository's positive logarithmic-density criterion do not cover the zero-density examples (4) and (28).

For $a_n = 2^{-n^2}$, consecutive logarithmic gaps are of order $\sqrt{(}z_n)$; for $a_n = 2^{-2^n}$, they are of order z_n. Neither satisfies our hypotheses, nor the local annular condition. Both, and arbitrary null sequences, remain outside our result. The proof also does not cover arbitrary flat germs, zero leading coefficients, or an uncontrolled $o(a^s)$ error. All sets are existence constructions; the finite tables can be astronomically large. The exact checker supplies algebraic and finite controls, not a computed high-measure avoiding set. The analytic theorem has a complete accompanying Lean replay for the stated hypotheses; this formalization is an independent kernel check of the supplied proof and is not a claim of external human peer review.

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[4] Y. Zhang, *Continuum power avoidance for bounded logarithmic gaps*, 2026. https://doi.org/10.5281/zenodo.23249747. The finite routing architecture is inherited from this research line.